

Priti Gurung

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Research Overview

I am a Computer Science PhD student specializing in computer vision and multimodal AI. My research focuses on improving vision language model better understand the physical world. Currently, I am extending the TVL(Tactile-Vision-Language) dataset with material-type labels, training tactile-only classifiers, and examining how LLM-generated annotations compare to human labels with a growing interest in active learning to make annotation more efficient. I have also worked on person re-identification, developing training-free methods to improve feature representations.

Education

Howard University

Ph.D. in Computer Science, GPA: 3.9/4.0

Washington, D.C.

Jan 2024 – Dec 2027 (Expected)

Tribhuvan University

B.Sc. in Computer Science and Information Technology, GPA: 3.4/4.0

Kathmandu, Nepal

Dec 2015 – Dec 2019

Research & Publications

- **“Risk Profiling using LLM and Few-shot Prompting Technique” *NeurIPS Women in Machine Learning Workshop 2024 (Accepted-Poster Presentation)*** : Proposed an automated system for risk assessment using LLM, and prompt engineering. Designed a scoring framework (0–10) to quantify risk across financial stability, cybersecurity, and reputation management, scalable to large volumes of social media data.
- **“LOOM: Rethinking Feature Refinement for Person Re-Identification ” (*Under submission*)**: Developed a plug-and-play, training-free feature enhancement method that refines embeddings via leave-one-out mean centering. Proposed Identity-Aware (IA-LOOM) and Cluster-Aware (CA-LOOM) variants to handle supervised and label-free settings, reducing intra-class variance and improving feature separability, leading to consistent improvement in mAP and Rank-1 accuracy with less computational overhead.
- **Extended TVL Dataset for Material Classification (*Ongoing*)** Extended the TVL (Touch-Vision-Language) dataset with material-type annotations labeled ; developed a tactile-only encoder-based classifier achieving 76.3% test accuracy , investigating LLM-annotated vs. human-labeled agreement rates to evaluate label quality. Can tactile information alone classify materials that are even visually ambiguous to humans?
- **“Getting the Right Attention:Body-Part Aware Prompting for Person Re-Identification” *AWS-Machine Learning University AI/ML Student Poster Session 2025 (Accepted - Poster session)*** : Proposed body-part-aware prompting to guide vision-language models, improving attention localization and robustness in person re-identification tasks.
- P. Gurung, J. Li, and D. B. Rawat, **“Towards neuro-symbolic reinforcement learning for trustworthy human-autonomy teaming,”** in Assurance and Security for AI-enabled Systems (J. D. Harguess, N. D. Bastian and T. L. Pace, eds.), vol. 13054, p. 1305407, International Society for Optics and Photonics, SPIE, 2024 (*Oral Presentation*)

Technical Skills

- **Programming:** Python, C++, C, SQL, MATLAB
- **Machine Learning & AI:** PyTorch, TensorFlow, Transformers, HuggingFace, Scikit-learn, OpenCV, CLIP, Vision-Language Models, Multimodal Learning (Vision–Language–Tactile), Representation Learning
- **GPU & Systems:** CUDA, GPU Acceleration (PyTorch), Parallel Computing, Linux
- **Robotics & Embodied AI:** Perception for manipulation, robot grasping, sensor fusion (vision + tactile), physical property reasoning (e.g., fragility)
- **Tools:** Git, Jupyter
- **Cloud:** AWS (EC2, S3)

Professional Experience

Research and Teaching Assistant

Howard University

Washington, DC

Jan 2024 – Present

- Investigating limitations of vision-only models in understanding physical and material properties(*Predicting object attributes such as: Fragile vs non-fragile, Soft vs rigid*)
- Studying how physical understanding improves robust decision-making in manipulation and interaction.
- Exploring dataset-driven approaches (e.g., Touch-Vision-Language Dataset) for multimodal alignment.
- Developed LOOM, a lightweight, training-free method to improve person re-identification by refining feature embeddings at inference time, without retraining or changing model architectures.
- Identified limitations in existing approaches, such as Pose2ID, including identity mixing and added computational cost, and designed a more stable and efficient alternative.
- **Teaching Assistant:** Structure of Programming Languages, Intro to Computer Science(III), Applied Data Science, Theory of Computation, Computer Organization(I), Introduction to AI

Data Business Analyst

Cedar Gate Technologies

Greenwich, CT

Apr 2022 – Dec 2023

- Interpreted and analyzed large-scale US healthcare datasets using statistical techniques, uncovering trends in utilization and costs to support value-based care analytics.
- Conducted data cleansing, transformation, and aggregation across multi-source systems, mapping disparate fields into a common schema and improving data consistency by 25%.
- Created 20+ technical documentation of system updates, presented progress to cross-functional teams, and coordinated with stakeholders to ensure alignment across data operations.

Software Engineer / Project Manager

Tej Solutions Pvt. Ltd.

Kathmandu, Nepal

Jun 2021 – Mar 2022

- Led the end-to-end delivery of a logistics platform, overseeing requirement elicitation, sprint planning, and release management, while contributing to core development in Django, React.js, and REST APIs.
- Coordinated with a team of 3 developers, 1 QA, and 1 designer, establishing Agile workflows and ensuring on-time delivery of web and mobile features for the client.
- Research and implemented delivery route and pricing optimization logic for a startup delivery app, reducing operational costs and improving scalability across locations.
- Developed the app of a financial reconciliation platform for internal use, automating file comparisons using Python and SQL, which cut manual auditing workload by 60%.

Associate Software Engineer

Integrated RCM Solution & Kathmandu Tech

Remote

Jan 2020 – Mar 2022

- Automated structured data synchronization workflows in Python/Excel macros, reducing manual intervention by 20%.
- Researched legal processes and designed form-based automation tools using no-code solutions, mapping user responses from questionnaires to structured legal documentation templates.

Honors & Leadership

- Member, *IEEE Computer Society*
- Member, *Computer and Robot Vision, IEEE Robotics and Automation Technical Committee*
- Member, *IEEE Robotics and Automation Technical Committee on Machine Learning for Automation*
- Member, *Society of Women Engineers, Women in Engineering, Rewriting the Code*
- Mentor and Graduate Student Judge, *BisonBytes Hackathon*, March 2025
- Reviewer, *Women in Machine Learning (WiML) @ NeurIPS*, 2024–2025
- Travel Fund Recipient: *WiML @ NeurIPS* (2024), *WiCV @ CVPR* (2025)
- Sub-Winner, *NASA Space Apps Challenge*, 2017
- AI Fellowship Scholar, *Fusemachines*, 2018
- First Runner-up, *Miss Mongol Nepal*, 2018
- Founder, *Miss Mongol Scholarship Foundation*, 2019
- Participation, *NVIDIA Agents for Impact - Hackathon at Howard University* 2025

Projects

- Facial Emotion Recognition System (2019): Present an approach based on Convolutional Neural Networks (CNN) for facial expression recognition. This was my undergrad thesis project.
- **Codepath - Applied AI Engineering Projects :**
 - Music Recommender Simulation–Built a content-based song recommender using weighted feature scoring (genre, mood, energy)
 - PawPal+ — Developed a pet care scheduling app with a priority-based greedy scheduler and conflict detection
 - The Mood Machine–Built an emotion classification simulation mapping input features to predicted mood states